scientific data

Check for updates

OPEN

DATA DESCRIPTOR

A Comprehensive Dataset for Word-Wheel Water Meter Reading Under Challenging Conditions

Shunyi Zhao[1]✉, Yibo Gao[1], Fei Liu[1], Yuxuan Zhang[2] & Jonghui Li[3]

We present a comprehensive dataset designed for segmentation, recognition, and classification tasks related to word-wheel type water meter reading. This dataset encompasses a wide range of real-world scenarios, including clear, blurry, reflective, and obstructed images, captured under various environmental conditions. As a result, it provides a robust benchmark for model training and evaluating. It contains over 50,000 water meter images, annotated with segmentation masks, recognition labels, and multi-hot encoded classification labels. These annotations facilitate the training of models for segmentation, recognition and multi-task classification, enabling them to address various challenges. Technical validation highlights the effectiveness and utility of the dataset in segmentation, recognition, and classification tasks across various challenge scenarios.

Introduction

Digital recognition in static images is a key area of computer vision, with application spanning multiple industries. These systems typically use image processing or segmentation models to isolate regions containing numbers, followed by transformations such as translation, rotation, and perspective adjustments to standardize the image. The preprocessed image is then passed to a number recognition model to be converted into digital data. Digital recognition has demonstrated notable success in practical applications, including license plate recognition in smart traffic systems and parking management[1,2], liquid level detection in laboratory and industrial environments[3,4], and text detection in document scanning systems[5–7]. These applications contribute to enhanced efficiency in various production processes.

Similarly, reading water meter data from images is also a form of digital recognition that plays a crucial role in resource monitoring. Water meters, essential tools for measuring water flow, are widely used across residential, commercial, and industrial sectors. In recent years, the development of Internet of Things technology has enabled the intelligence of smart city infrastructure, improved environmental monitoring, and optimized energy management. These advancements are driving urban services toward greater efficiency and sustainability[8,9]. Consequently, the rise of smart water meters, which offer automated, accurate, and remote readings with direct data uploads[10,11], has attracted significant research interest. These meters reduce manual labor and minimize human error, improving the efficiency of water management systems. Despite these advantages, the high cost of smart meters has limited their widespread adoption, especially in smaller towns and less developed regions. In these areas, traditional water meters—featuring pointer-type or word-wheel-type counters—remain prevalent. These meters use mechanical gears driven by water flow to measure consumption, and their readings are typically collected manually by utility companies. This process is not only labor-intensive but also prone to inaccuracies, underscoring the need for automation. Automating the reading of traditional water meters offers a cost-effective solution to enhance efficiency, reduce errors, and support the broader development of smart infrastructure. To address this challenge, we have developed a comprehensive dataset specifically tailored for traditional word-wheel type water meters, providing a solid foundation for advancing automated meter reading technologies.

Challenges. Automatically reading word-wheel water meters through computer vision in diverse scenarios presents substantial challenges. Unlike applications such as license plate recognition or liquid level detection, water meter reading is especially susceptible to a variety of environmental and technical factors. Water meters are

[1]Key Laboratory of Advanced Process Control for Light Industry, Ministry of Education, Jiangnan University, Wuxi, China. [2]Department of Management Science and Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing, China. [3]Zhejiang Bainuo Digital Intelligence Environmental Technology Co., Ltd, Hangzhou, China. ✉e-mail: shunyi@jiangnan.edu.cn



Fig. 1 Representative images of each classified type within the dataset: (**a**) Clear; (**b**) dial-stained or covered; (**c**) blurry; (**d**) reflective or dark.

often installed in outdoor or industrial environments, where the dial may be obscured by soil, debris, or vegetation, and lighting conditions are frequently suboptimal. Poor illumination can result in dim or shadowed images, making it difficult to capture clear and unobstructed views of the meter.

Additionally, image acquisition introduces further complexity. Issues such as camera misfocus, angle misalignment, and reflective glare from the meter surface often result in blurred or distorted images, which degrade recognition accuracy. Subtle variations in ambient lighting, shadows, or reflections can also significantly impact the reliability of digital recognition systems. These combined factors make automatic reading of traditional water meters a complex problem to solve. To address these challenges, a robust and representative dataset is crucial. Such a dataset would support the development of more resilient recognition methods capable of performing reliably under varied conditions.

Related work. In this paper, we present a dataset specifically designed for word-wheel water meter reading. Currently, datasets used to train water meter reading models remain limited. The publicly available datasets often have issues such as a small sample size, high image similarity, and inadequate representation of diverse real-world scenarios[12]. The dataset contains over 50,000 water meter images, all captured in diverse real-world scenarios[13]. This collection includes not only images with clear, fully visible dials but also images exhibiting common challenges encountered in manual meter reading, such as stained dials and blurred images. Representative images of each type are shown in Fig. 1.

The dataset is carefully curated to include only valid images, which have been standardized in terms of size and format. Separate segmentation and recognition labels are provided for image segmentation and number recognition tasks. Due to the diversity of image types, multi-hot encoding classification labels are also included. These labels enable the training of classification models that can recognize a range of real-world factors. By refining model performance across these varied scenarios, the dataset enhances the adaptability of water meter reading solutions in practical applications.

This dataset is publicly available, providing a valuable resource for developing water meter reading solutions in complex real-world conditions. The second section of this paper outlines the image collection and dataset construction process, while the third section provides detailed usage instructions, including the directory structure and dataset composition statistics. Finally, the last section evaluates the performance of existing segmentation, recognition, and classification algorithms on this dataset, summarizing their effectiveness across different challenging scenarios.

Methods

The images in this dataset were collected from the actual manual meter reading process in Hangzhou City, the second-largest city in the Yangtze River Delta region of China. The city's underground water infrastructure is complex, comprising many traditional water meters of varying ages. As such, the collected images of these traditional water meters are very representative of real-world conditions. The model trained with this dataset can be easily extended to other cities with similar infrastructure. In constructing this dataset, we took into account various real-world challenges based on the experiences of frontline meter reading workers. The water meter status in the dataset is highly varied, with different degrees of coverage and visual distortion resulting from factors such as shooting angles and obstructions.

The task of reading word-wheel type water meters from images involves two distinct steps: detection and recognition. Each of these steps requires different models and separately constructed datasets. The following sections outline the construction process for the datasets used in detection and recognition.

Detection set construction. In the detection step, the primary goal is to detect and segment the water meter's reading area within each image. The dataset construction process involves the following three key steps:

1) Image Screening and Size Standardization: The collected images vary in size and quality, with some containing incomplete or missing water meter reading areas. Invalid images were manually removed, and all remaining images were resized to a uniform dimension of 512×512 pixels.

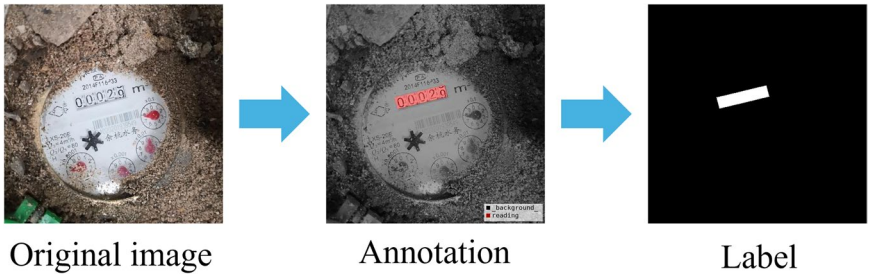


Fig. 2 Segmentation label annotation example.

2) Segmentation Label Annotation: Using the *LabelMe* tool, each image was annotated with a quadrilateral bounding box around to the reading area. An example of the segmentation label annotation is shown in Fig. 2. The annotations were then converted into binary masks, where the reading area is represented in white, and the background is black.
3) Classification Label Annotation: In addition to the foreground segmentation labels, we also provide classification labels to represent complex real-world scenarios and assist in water meter reading recognition. Since a single image may exhibit multiple scenarios simultaneously—such as being both blurry and reflective—multi-hot encoding is used to represent these conditions. Figure 3 shows the arrangement of the classification labels.

The first column contains the file name, which also serves as the index for the classification label. The subsequent columns indicate whether the image belongs to the corresponding type of scenario. These columns are ordered as follows: *clear, blurry, dial-stained, soil-covered, dark, reflective*, and *six-digit*. If an image corresponds to a particular type, the value is marked as 1; otherwise, it is marked as 0. For example, an image that is both blurry and reflective would be represented by the binary string 010001, while a clear but dark image would be represented as 100010. These binary strings are stored in a CSV file as classification labels.

Through the above steps, the detection set construction is completed. For water meter images captured under varying real-world scenarios, the classification label serves as a summarize description of scenario. By providing these labels, we aim to improve model training based on environmental features, leading to more accurate water meter reading methods. Specifically, this approach offers several advantages: The original images can be adjusted based on their classification labels. For instance, image enhancement or restoration techniques can be applied according to the identified scenario, thereby improving model recognition efficiency. These classification labels also allow for adjustments in the distribution of image types during model training. This helps ensure balanced model performance by controlling the proportion of each image type in the training set and enables the development of specialized models tailored to specific scenarios. During testing, model performance can be evaluated separately for each scenario. This approach helps identify which environmental factors most impact model accuracy, allowing for targeted improvements to enhance model robustness across diverse conditions.

Recognition set construction. In the recognition step, the local images of water meter reading generated during segmentation are processed for digital recognition. The construction process involves the following steps:

1) Image Screening: Since water meter readings consist of five or six digits, the range of potential readings is limited. Therefore, it is unnecessary to include all segmentation results in the recognition step. A representative selection of images is sufficient, including clear readings as well as challenging cases such as blurred digits, half-word readings, and so on.
2) Projection Transformation: Segmented images often appear as irregular shapes. To standardize the input for recognition algorithms, all segmented areas were transformed into 60×200 pixels rectangular images in JPG format.
3) Label Annotation: The corresponding labels used in the recognition step are stored in a CSV file, the first column contains the image file names as index, the second column contains the expected five-digit or six-digit recognition results, and columns three through five contain multi-hot encoded labels. These additional labels classify scenarios encountered during recognition step, including *reversed, half-word* and *six-digit*, following the same principle as the classification labels used in the detection step. Figure 4 provides a structural diagram of the CSV file used for the recognition task.

Data Records

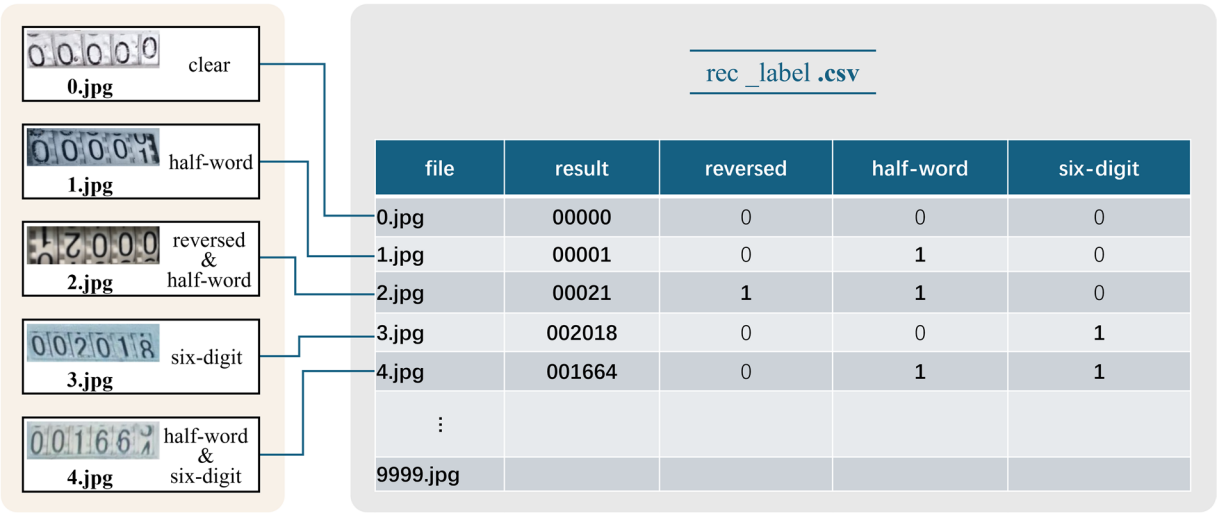
Our dataset is now available for download at *Dryad* (https://doi.org/10.5061/dryad.7d7wm3860).

Directory structure. After downloading and unzipping the dataset, root directory contains two primary subdirectories: one for detection and another for recognition. The detection directory is intended for training the model to detect and segment the reading area in water meter images, containing train set and test set. The directory structure of the train and test set is the same, both include the following three files:



file	clear	blurry	dial-stained	soil-covered	dark	reflective	six-digit
0.png	1	0	0	0	0	0	0
1.png	0	0	0	0	0	1	0
2.png	0	1	1	0	0	0	0
⋮							
9999.png							

Fig. 3 Classification label annotation example.



file	result	reversed	half-word	six-digit
0.jpg	00000	0	0	0
1.jpg	00001	0	1	0
2.jpg	00021	1	1	0
3.jpg	002018	0	0	1
4.jpg	001664	0	1	1
⋮				
9999.jpg				

Fig. 4 Recognition label annotation example.

1) *det_img*: This subdirectory contains images to be segmented. The train set contains 32,714 images which are named sequentially from *train0.png* to *train32713.png*, while test set contains 13,398 images (*test0.png* to *test13397.png*). All images are in PNG format and have a resolution of 512 × 512 pixels.
2) *seg_label*: This subdirectory contains the segmentation labels. Each label file has the same name as its corresponding image in the *det_img* directory. The labels are also in PNG format and are the same size as the images.
3) *class_label.csv*: This CSV file contains scenario classification labels for each image. The first column contains the file name corresponding to image in the *det_img* directory as index. Columns two through seven contain multi-hot encoded binary strings representing the scenarios of this image.

The recognition directory is used to train the digit recognition model. The images in this directory are categorized into two distinct subdirectories based on the number of digits: one for 5-digit readings and the other for 6-digit readings. Each subdirectory contains a training set and a test set. These sets include the following files:

1) *rec_img*: This subdirectory contains local images of water meter readings. The 5-digit dataset contains 11,867 training images (*train0.jpg* to *train11866.jpg*) and 2,000 test images (*test0.jpg* to *test1999.jpg*). The 6-digit dataset contains 2,555 training images (*train0.jpg* to *train2554.jpg*) and 400 test images (*test0.jpg* to *test399.jpg*). All images are in JPG format with a resolution of 60 × 200 pixels.
2) *rec_label.csv*: This CSV file includes the labels used during the recognition step. The first column contains the file name corresponding to image in the *rec_img* directory. The second column contains the 5-digit or 6-digit recognition label of this image. Columns three through five contain classification labels of this image.

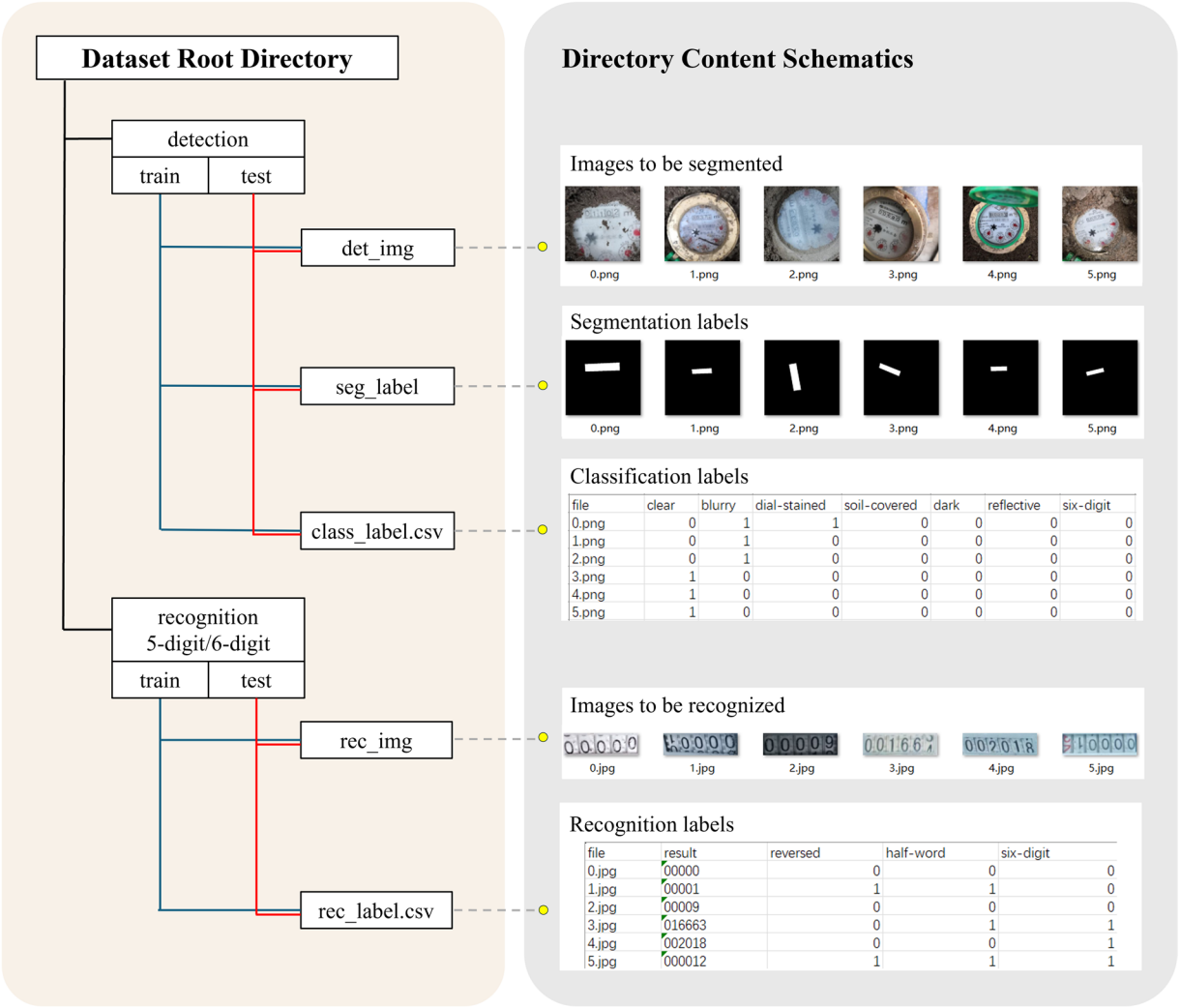


Fig. 5 Dataset directory structure overview.

Directory	Size (pixels)	Format	Storage (KB)
det_img	512 × 512	PNG	40.0
seg_label	512 × 512	PNG	40.0
rec_img	60 × 200	JPG	12.0

Table 1. Statistics of image size, format and storage.

Figure 5 provides an overview of the dataset directory structure, with schematic diagrams for each directory or file included for reference. Table 1 summarizes the size, format, and storage details of each directory containing images.

Composition. To ensure the trained model is robust and adaptable to real-world practice, this dataset is composed of specific proportions representing various challenging scenarios. In the detection set, images with dial stains and soil coverage constitute the majority, reflecting the common occurrence of water meters installed in wells or underground pipelines where soil and debris often accumulate, obscuring or partially covering the readings. Blurry images are prevalent, caused by focus issues, low resolution, or obstructions during image acquisition. Dark and reflective conditions are common, often due to inadequate lighting in underground installations or improper lighting and camera angles that cause glare. Additionally, due to different models of water meters, a small portion of the dataset includes images with six-digit readings.

In the recognition set, the dataset includes three specific scenarios that are critical for handling complex situations: reversed readings, half-word, and six-digit readings. These cases account for situations where the segmentation results are reversed, where the counter advances to an intermediate position causing partial digits to appear, or where the segmentation is derived from a six-digit model water meter. The detailed statistics for these categories are shown in Table 2. This composition ensures that the trained model can be effectively applied in practice.

Detection set					Recognition set				
Type	Count				Type	Count			
	Train		Test			Train		Test	
clear	6902	21%	3025	23%	reversed (5)	5709	48%	629	31%
blurry	2620	8%	1260	9%	half-word (5)	4235	36%	355	18%
dial-stained	13662	42%	4685	35%	Total (5)	11867		2000	
soil-covered	12349	38%	4442	33%		13867			
dark	728	2%	300	2%					
reflective	1660	5%	570	4%	reversed (6)	1067	42%	150	38%
six-digit	2262	7%	955	7%	half-word (6)	997	39%	151	38%
Total	32714		13398		Total (6)	2555		400	
	46112					2955			

Table 2. Detection set and Recognition set composition statistics. In the recognition set, the five-digit reading is marked with (5) and the six-digit reading is (6).



Fig. 6 Schematic diagram for determining the effectiveness of segmentation.

Technical Validation

To evaluate the utility and robustness of dataset construction, we use several popular segmentation and recognition algorithms separately on the detection set and recognition set. The technical validation was conducted on a system equipped with CPU and GPU. The environment setup included PyTorch as the deep learning framework, with CUDA for GPU acceleration.

Detection set validation. We selected four widely used segmentation algorithms: UNet, DeepLabV3+, PSPNet, and SegFormer. Each algorithm was chosen for its unique strength in handling complex visual tasks:

1) UNet is a well-established architecture known for its efficient segmentation in medical and industrial images. Its symmetric encoder-decoder structure allows for precise localization and segmentation of small objects[14].
2) DeepLabV3 + builds upon the DeepLab family with advanced dilated convolutions and spatial pyramid pooling, making it particularly effective in segmenting complex and varied image textures, such as those in our dataset[15].
3) PSPNet utilizes a pyramid pooling module that captures global context at multiple scales, which helps in recognizing objects that vary widely in scale and placement[16].
4) SegFormer is a transformer-based model that combines the strengths of transformers for capturing long-range dependencies with lightweight, efficient architecture, making it suitable for high-resolution image segmentation while maintaining computational efficiency[17].

Several key metrics were used to evaluate and compare the performance of the models: Valid rate, Precision, Recall, and F1-Score, each providing the effects in different aspects. The ratio of the total number of valid predictions to the total number of test set is called valid rate. We overlap the predicted labels generated during the testing phase on the original image and manually check whether the label fully covers the reading area of the water meter, without missing digits or covering excess, such predicted results are recorded as valid segmentation. If the label area misses digits, covers extra irrelevant areas, or fails to detect the reading area, it will be considered invalid. The schematic diagram is shown in Fig. 6. Additionally, Precision measures the accuracy in identifying relevant areas, Recall measures the ability to capture all relevant areas. F1-Score is the harmonic mean of Precision and Recall[13].

Model	Train time	Model size	Valid ratio	Precision	Recall	F1-Score	Inference time
UNet	32 min	87 MB	**98.60%**	88.58%	97.87%	92.99%	1.28 ms
DeepLabV3+	106 min	160 MB	98.20%	91.05%	97.15%	**94.00%**	1.58 ms
PSPNet	**31 min**	93 MB	96.80%	90.40%	96.65%	93.42%	**0.86 ms**
SegFormer	32 min	**14 MB**	**98.60%**	89.76%	97.57%	93.50%	2.04 ms

Table 3. Detection set validation result.

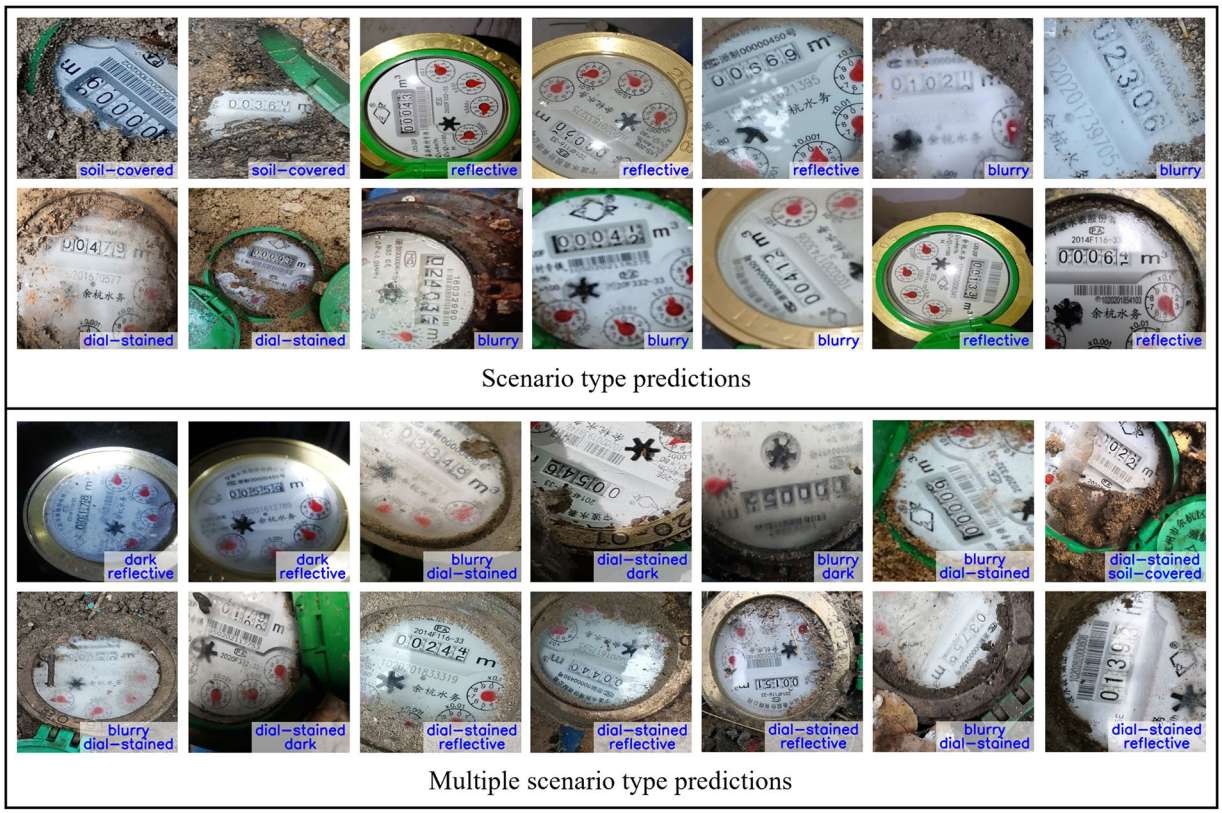


Fig. 7 Scenario type prediction visualization for classification validation.

Table 3 presents the validation result of detection set using different methods. The top of Fig. 9 shows the F1-Score curves of each model over epochs. All models converged rapidly and achieved high prediction accuracy, demonstrating the effectiveness of detection set for water meter reading area segmentation.

Classification validation. In addition to the segmentation task, we also conducted technical verification on the classification labels, evaluating their effectiveness from two perspectives. The first aspect is to determine whether these classification labels can successfully complete the classification task. The second aspect is to assess whether the classification labels can assist in the segmentation task, providing additional context for more accurate results in challenging environments.

We used ResNet to validate the classification labels. Figure 7 visualizes the prediction results on the test set. Most of the pictures are classified correctly. The middle of Fig. 9 shows the classification F1-Score curves over epochs under different scenarios. The results verify the validity of the classification labels.

Table 4 presents the segmentation F1-Score of each model and the final classification F1-Score of ResNet under different scenarios. When the classification label is used to assist in the segmentation task, we observe that all the four models perform best under clear conditions, with varying degrees of decrease in accuracy in different challenging environments. Specifically, the accuracy drops significantly in dark conditions, highlighting the difficulties posed by poor lighting. These findings suggest that the use of classification labels to aid in segmentation is a viable approach. Compared to traditional segmentation methods, the classification labels can provide more scenario information to enhance robustness of the model in complex environments.

Recognition set validation. We selected several digit recognition networks: CNN, VGG16, ResNet, and DenseNet. Models were trained and tested on the recognition set to evaluate its effectiveness in supporting accurate digit recognition.

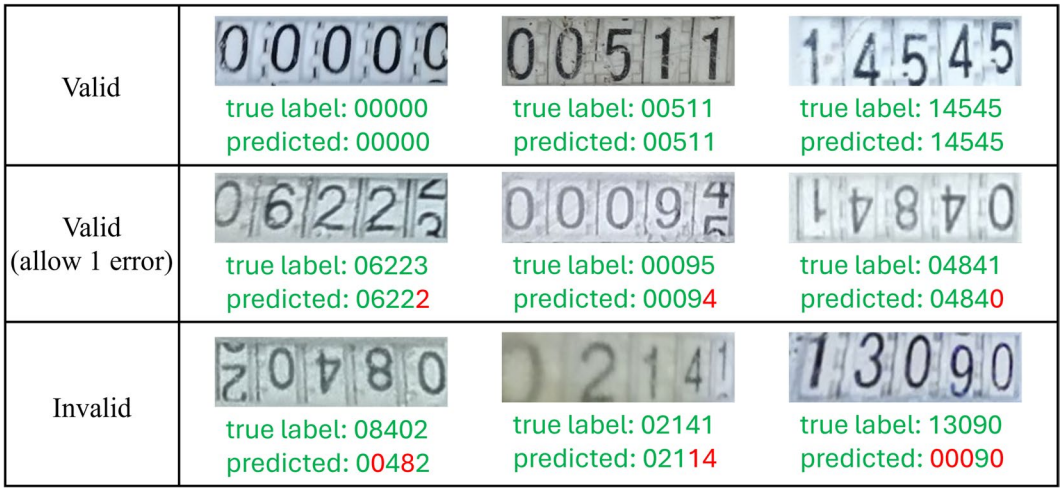


Fig. 8 Schematic diagram for correct, allow 1 error, and incorrect digit recognition.

F1-Score	Model	Type					
		clear	blurry	dial-stained	soil-covered	dark	reflective
Segmentation	UNet	**93.18%**	89.00%	91.69%	91.52%	**87.92%**	89.69%
	DeepLabV3+	**93.58%**	89.25%	91.88%	91.17%	**87.99%**	89.78%
	PSPNet	**92.74%**	83.12%	89.63%	88.96%	**75.34%**	86.19%
	SegFormer	**93.35%**	90.42%	92.01%	91.91%	**88.44%**	90.12%
Classification	ResNet	—	83.53%	89.64%	87.72%	93.18%	88.10%
	Average	88.43%					

Table 4. The segmentation F1-Score of each model and the classification F1-Score of ResNet under different scenarios.

Model	Train time	Model size	Accuracy	Accuracy (allow 1 error)	Inference time
CNN	**512 s**	**44 MB**	87.05%	97.15%	**0.03 ms**
VGG16	2054 s	512 MB	94.30%	99.10%	0.12 ms
ResNet	520 s	**44 MB**	97.85%	**99.80%**	0.06 ms
DenseNet	1104 s	55 MB	**98.15%**	**99.80%**	0.41 ms

Table 5. Recognition set validation result.

1) CNNs are fundamental architectures in computer vision, which were chosen for their simplicity, computational efficiency, and proven effectiveness in foundational image recognition tasks, serving as a baseline for comparison[18].
2) VGG16 is a deep convolutional network, augmenting the depth and complexity of the network, consisting of 16 layers. It enables to extract detailed features while preventing overfitting combined with the use of smaller filters[19].
3) ResNet and DenseNet both have advantages to capture complex feature representations aligns well with the challenges posed by our dataset, particularly in handling blurred or partially obscured images[20,21].

Recognition accuracy was used to evaluate the performance of each model. We separately analyzed all recognition results by categorizing them into correct, one-error, and invalid cases. Based on these cases, we calculated both the standard accuracy and the accuracy under the condition of allowing one error. These cases are visually represented in Fig. 8.

Table 5 shows the performance of models. CNN has the smallest model size and the shortest train and inference time with its simple network structure. Therefore, its accuracy is obviously lower compared to the other models. VGG16 improves accuracy but requires a larger model size and longer training time. ResNet strikes a balance, delivering high accuracy with a smaller model size and shorter inference time. DenseNet achieves the highest accuracy, but its inference time is about seven times longer than ResNet. The bottom of Fig. 9 shows the accuracy curves of each recognition model on the test set over epochs.

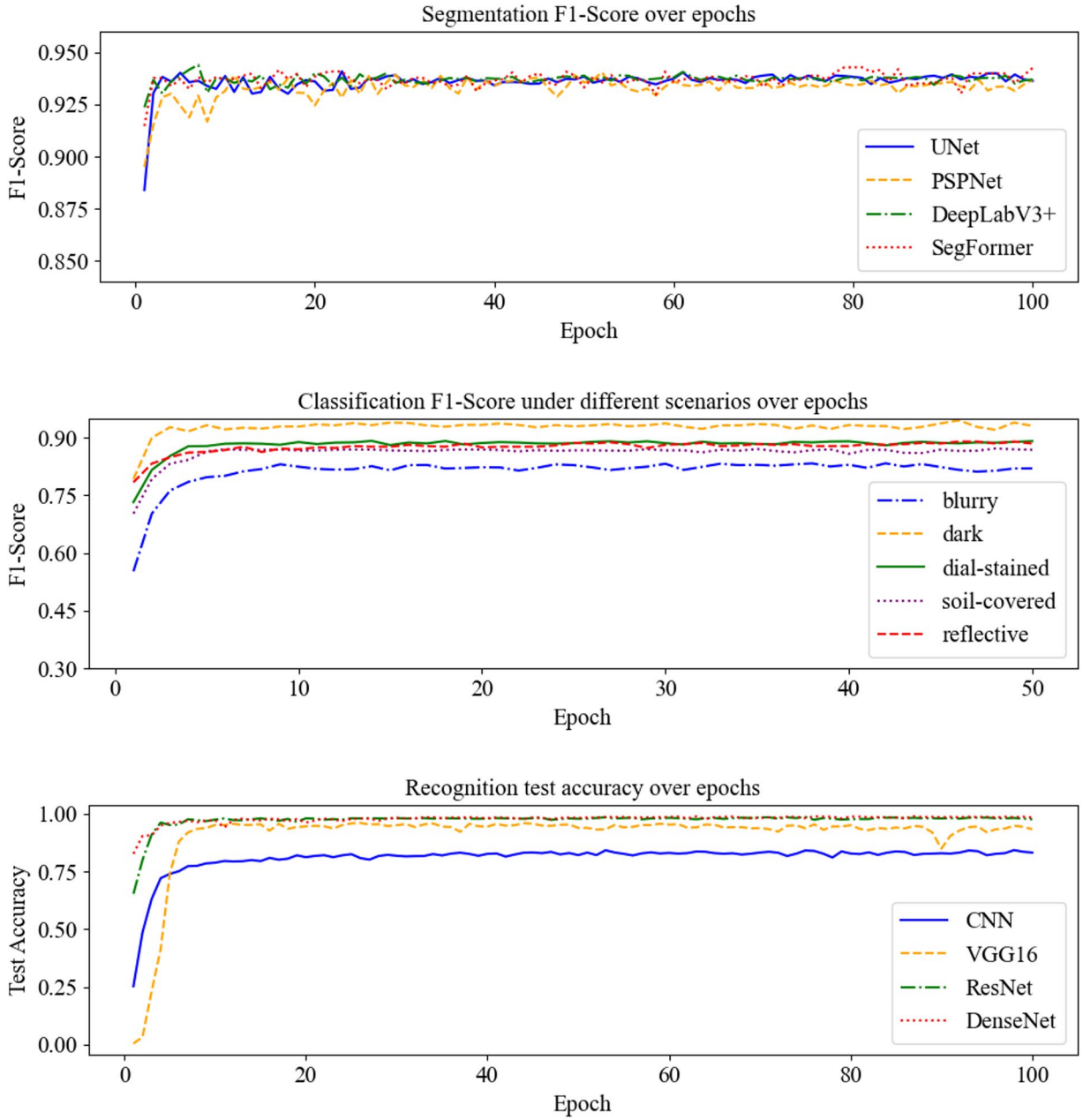


Fig. 9 Segmentation F1-Score of each model (detection set validation), Classification F1-Score under different scenarios (classification validation), and Recognition test accuracy of each model (recognition set validation) with epochs increasing.

Data availability

The dataset is publicly available at the Dryad repository under the https://doi.org/10.5061/dryad.7d7wm3860[22].

Code availability

The code used in this paper is available at https://github.com/100000000Bo/water_meter_dataset_technical_validation_python_project.git. The repository provides data preprocessing code and technical validation code. The data preprocessing code covers image preprocessing, annotation, and dataset construction. The technical validation code supports water meter classification, recognition, and segmentation tasks, including training and testing scripts for multiple baseline deep learning models. The required Python environments for data preprocessing and validation are specified in the provided *env_preprocess.yml* and *env_validation.yml* files. Detailed instructions for code usage are provided in the accompanying documentation files, *README_Preprocess.md* and *README_Validation.md*.

Received: 25 March 2025; Accepted: 3 February 2026;
Published online: 19 February 2026

References

1. Chang, S.-L., Chen, L.-S., Chung, Y.-C. & Chen, S.-W. Automatic license plate recognition. *IEEE Transactions on Intelligent Transportation Systems* **5**, 42–53 (2004).
2. Du, S., Ibrahim, M., Shehata, M. & Badawy, W. Automatic license plate recognition (ALPR): A state-of-the-art review. *IEEE Transactions on Circuits Systems for Video Technology* **23**, 311–325 (2012).
3. Syed, J., Takriti, A. & Damindarov, R. Computer vision and deep learning enabled real-time liquid level detection and measurement in transparent containers. *International Conference on Industrial Engineering, Applications and Manufacturing*. 711–716 (2024).
4. Narayan, A., Charan, D., Saji, S. E., Fang, T. & Saniie, J. A PyTorch-based deep learning approach for enhanced liquid level detection in industrial environments. *IEEE International Conference on Electro Information Technology*. 562–566 (2024).
5. Epshtein, B., Ofek, E. & Wexler, Y. Detecting text in natural scenes with stroke width transform. *IEEE Computer Society Conference on Computer Vision and Pattern Recognition*. 2963–2970 (2010).
6. Nathiya, N. & Pradeepa, K. Optical character recognition for scene text detection, mining and recognition. *IEEE International Conference on Computational Intelligence and Computing Research*. 1–4 (2013).
7. Baek, Y., Lee, B., Han, D., Yun, S. & Lee, H. Character region awareness for text detection. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 9365–9374 (2019).
8. Zanella, A., Bui, N., Castellani, A., Vangelista, L. & Zorzi, M. Internet of things for smart cities. *IEEE Internet of Things Journal* **1**, 22–32 (2014).
9. Brundu, F. G. *et al*. IoT software infrastructure for energy management and simulation in smart cities. *IEEE Transactions on Industrial Informatics* **13**, 832–840 (2016).
10. Zapata-Sierra, A. J., Salmerón-Manzano, E., Alcayde, A., Zapata-Castillo, M. L. & Manzano-Agugliaro, F. The scientific landscape of smart water meters: A Comprehensive Review. *Water* **16**, 113 (2023).
11. Hsia, S. C., Wang, S.-H. & Hsu, S.-W. Smart water-meter wireless transmission system for smart cities. *IEEE Consumer Electronics Magazine* **10**, 83–88 (2020).
12. Zhu, G., Zhao, Q., Zhang, Z., Huang, Q. & Cheng, M. Character region extraction of wheel water meter based on object detection. *Flow Measurement Instrumentation* **100**, 102733 (2024).
13. Salari, A., Djavadifar, A., Liu, X. & Najjaran, H. Object recognition datasets and challenges: A review. *Neurocomputing* **495**, 129–152 (2022).
14. Ronneberger, O., Fischer, P. & Brox, T. U-net: Convolutional networks for biomedical image segmentation. *Medical Image Computing and Computer-Assisted Intervention International Conference*. 234–241 (2015).
15. Chen, L.-C., Zhu, Y., Papandreou, G., Schroff, F. & Adam, H. Encoder-decoder with atrous separable convolution for semantic image segmentation. *Proceedings of the European Conference on Computer Vision*. 801–818 (2018).
16. Zhao, H., Shi, J., Qi, X., Wang, X. & Jia, J. Pyramid scene parsing network. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2881–2890 (2017).
17. Xie, E. *et al*. SegFormer: Simple and efficient design for semantic segmentation with transformers. *Advances in Neural Information Processing Systems* **34**, 12077–12090 (2021).
18. Chauhan, R., Ghanshala, K. K. & Joshi, R. Convolutional neural network (CNN) for image detection and recognition. *International Conference on Secure Cyber Computing and Communication (ICSCCC)*. 278–282 (2018).
19. Simonyan, K. & A. Zisserman. Very deep convolutional networks for large-scale image recognition. *Proceedings of the 3rd International Conference on Learning Representations*. (2015).
20. He, K., Zhang, X., Ren, S. & Sun, J. Deep residual learning for image recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 770–778 (2016).
21. Huang, G., Liu, Z., Van Der Maaten, L. & Weinberger, K. Q. Densely connected convolutional networks. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 4700–4708 (2017).
22. Zhao, S. *et al*. A comprehensive dataset for word-wheel water meter reading under challenging conditions. *Dryad*. https://doi.org/10.5061/dryad.7d7wm3860 (2026).

Acknowledgements

This work was supported by the National Natural Science Foundation of China under Grant No. 62373167 and by the Natural Science Foundation of Jiangsu Province under Grant No. BK20241398.

Author contributions

Professor Shunyi Zhao is responsible for the construction of the dataset, as well as the review and editing of the article. Yibo Gao handles label annotation, technical validation, and writing the original draft. Professor Fei Liu contributes to data analysis, as well as the article review and editing. Professor Yuxuan Zhang is responsible for data analysis and ensuring quality control. Mr. Jonghui Li oversees data collection.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to S.Z.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026